

Laboratory work #2 DDL section

pgAdmin 4

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Query

```
1 CREATE TABLE airline_info (  
2     airline_id INT NOT NULL PRIMARY KEY,  
3     airline_code VARCHAR(30) NOT NULL,  
4     airline_name VARCHAR(50) NOT NULL,  
5     airline_country VARCHAR(50) NOT NULL,  
6     created_at TIMESTAMP NOT NULL,  
7     updated_at TIMESTAMP NOT NULL,  
8     info VARCHAR(50) NOT NULL  
9 );  
10  
11 CREATE TABLE airport (  
12     airport_id INT NOT NULL PRIMARY KEY,  
13     airport_name VARCHAR(50) NOT NULL,  
14     country VARCHAR(50) NOT NULL,  
15     state VARCHAR(50) NOT NULL,  
16     city VARCHAR(50) NOT NULL,  
17     created_at TIMESTAMP NOT NULL,  
18     updated_at TIMESTAMP NOT NULL  
19 );  
20  
21 CREATE TABLE passengers (  
22     passenger_id INT NOT NULL PRIMARY KEY,  
23     first_name VARCHAR(50) NOT NULL,  
24     last_name VARCHAR(50) NOT NULL,  
25     date_of_birth DATE NOT NULL,  
26     gender VARCHAR(50) NOT NULL,  
27     country_of_citizenship VARCHAR(50) NOT NULL,  
28     country_of_residence VARCHAR(50) NOT NULL,  
29     passport_number VARCHAR(20) NOT NULL,  
30     created_at TIMESTAMP NOT NULL,  
31     updated_at TIMESTAMP NOT NULL  
32 );
```

Query History

Scratch Pad

Data Output Messages Notifications

CREATE TABLE

Query returned successfully in 61 msec.

Total rows: Query complete 00:00:00.061

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Query

```
1 CREATE TABLE flights (  
2     flight_id INT NOT NULL PRIMARY KEY,  
3     sch_departure_time TIMESTAMP NOT NULL,  
4     sch_arrival_time TIMESTAMP NOT NULL,  
5     departing_airport_id INT NOT NULL,  
6     arriving_airport_id INT NOT NULL,  
7     departing_gate VARCHAR(50) NOT NULL,  
8     arriving_gate VARCHAR(50) NOT NULL,  
9     airline_id INT NOT NULL,  
10    act_departure_time TIMESTAMP NOT NULL,  
11    act_arrival_time TIMESTAMP NOT NULL,  
12    created_at TIMESTAMP NOT NULL,  
13    updated_at TIMESTAMP NOT NULL  
14 );  
15  
16 CREATE TABLE booking (  
17     booking_id INT NOT NULL PRIMARY KEY,  
18     flight_id INT NOT NULL,  
19     passenger_id INT NOT NULL,  
20     booking_platform VARCHAR(50) NOT NULL,  
21     created_at TIMESTAMP NOT NULL,  
22     updated_at TIMESTAMP NOT NULL,  
23     status VARCHAR(50) NOT NULL,  
24     price DECIMAL(7,2) NOT NULL  
25 );  
26  
27 CREATE TABLE booking_flight (  
28     booking_flight_id INT NOT NULL PRIMARY KEY,  
29     booking_id INT NOT NULL,  
30     flight_id INT NOT NULL,  
31     created_at TIMESTAMP NOT NULL,  
32     updated_at TIMESTAMP NOT NULL  
33 );
```

Query History

Scratch Pad

Data Output Messages Notifications

CREATE TABLE

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Dashboard x Properties x SQL x Statistics x Dependencies x Dependents x Processes x airport/postgres@PostgreSQL 18* x

airport/postgres@PostgreSQL 18

No limit

Query Query History

```
1 CREATE TABLE baggage (
2     baggage_id INT NOT NULL PRIMARY KEY,
3     weight_in_kg DECIMAL(4,2) NOT NULL,
4     created_at TIMESTAMP NOT NULL,
5     updated_at TIMESTAMP NOT NULL,
6     booking_id INT NOT NULL
7 );
8
9 CREATE TABLE baggage_check (
10     baggage_check_id INT NOT NULL PRIMARY KEY,
11     check_result VARCHAR(50) NOT NULL,
12     created_at TIMESTAMP NOT NULL,
13     updated_at TIMESTAMP NOT NULL,
14     booking_id INT NOT NULL,
15     passenger_id INT NOT NULL
16 );
17
18 CREATE TABLE boarding_pass (
19     boarding_pass_id INT NOT NULL PRIMARY KEY,
20     booking_id INT NOT NULL,
21     seat VARCHAR(50) NOT NULL,
22     boarding_time TIMESTAMP NOT NULL,
23     created_at TIMESTAMP NOT NULL,
24     updated_at TIMESTAMP NOT NULL
25 );
26
27 CREATE TABLE security_check (
28     security_check_id INT NOT NULL PRIMARY KEY,
29     check_result VARCHAR(20) NOT NULL,
30     created_at TIMESTAMP NOT NULL,
31     updated_at TIMESTAMP NOT NULL,
32     passenger_id INT NOT NULL
33 );
```

Data Output Messages Notifications

CREATE TABLE

Query returned successfully in 67 msec.

Total rows: Query complete 00:00:00.067

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 - airport
 - baggage
 - baggage_check
 - boarding_pass
 - booking
 - booking_flight
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 - security_check
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Dashboard x Properties x SQL x Statistics x Dependencies x Dependents x Processes x airport/postgres@_ x

Name	Owner	Partitioned table?
☐ airline_info	postgres	☐
☐ airport	postgres	☐
☐ baggage	postgres	☐
☐ baggage_check	postgres	☐
☐ boarding_pass	postgres	☐
☐ booking	postgres	☐
☐ booking_flight	postgres	☐
☐ flights	postgres	☐
☐ passengers	postgres	☐
☐ security_check	postgres	☐

Task 2, 3

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Dashboard | Properties | SQL | Statistics | Dependencies | Dependents | Processes | airport/postgres@PostgreSQL 18*

Query Query History

```
1 SELECT table_name, column_name, is_nullable
2 FROM information_schema.columns
3 WHERE table_schema = 'public'
4 ORDER BY table_name;
```

Data Output Messages Notifications

	table_name	column_name	is_nullable
	name	name	character varying (3)
1	airline_info	updated_at	NO
2	airline_info	airline_id	NO
3	airline_info	airline_code	NO
4	airline_info	airline_name	NO
5	airline_info	airline_country	NO
6	airline_info	info	NO
7	airline_info	created_at	NO
8	airport	created_at	NO
9	airport	airport_id	NO
10	airport	airport_name	NO
11	airport	country	NO
12	airport	state	NO
13	airport	city	NO
14	airport	updated_at	NO
15	baggage	baggage_id	NO
16	baggage	weight_in_kg	NO
17	baggage	created_at	NO
18	baggage	updated_at	NO
19	baggage	booking_id	NO
20	baggage_ch...	check_result	NO
21	baggage_ch...	created_at	NO
22	baggage_ch...	updated_at	NO
23	baggage_ch...	passenger_id	NO
24	baggage_ch...	baggage_check_id	NO
25	baggage_ch...	booking_id	NO

Total rows: 71 Query complete 00:00:00.278

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Dashboard | Properties | SQL | Statistics | Dependencies | Dependents | Processes | airport/postgres@PostgreSQL 18*

Query Query History

```
1 SELECT table_name, column_name, is_nullable
2 FROM information_schema.columns
3 WHERE table_schema = 'public'
4 ORDER BY table_name;
```

Data Output Messages Notifications

	table_name	column_name	is_nullable
	name	name	character varying (3)
26	boarding_pa...	created_at	NO
27	boarding_pa...	boarding_time	NO
28	boarding_pa...	booking_id	NO
29	boarding_pa...	sest	NO
30	boarding_pa...	boarding_pass_id	NO
31	boarding_pa...	updated_at	NO
32	booking	booking_platform	NO
33	booking	status	NO
34	booking	price	NO
35	booking	created_at	NO
36	booking	booking_id	NO
37	booking	flight_id	NO
38	booking	passenger_id	NO
39	booking	updated_at	NO
40	booking_flight	updated_at	NO
41	booking_flight	created_at	NO
42	booking_flight	booking_flight_id	NO
43	booking_flight	booking_id	NO
44	booking_flight	flight_id	NO
45	flights	arriving_gate	NO
46	flights	flight_id	NO
47	flights	sch_departure_time	NO
48	flights	sch_arrival_time	NO
49	flights	departing_airport_id	NO
50	flights	arriving_airport_id	NO

Total rows: 71 Query complete 00:00:00.278

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Dashboard Properties SQL Statistics Dependencies Dependents Processes airport/postgres@PostgreSQL 18*

Query Query History

```
1 SELECT table_name, column_name, is_nullable
2 FROM information_schema.columns
3 WHERE table_schema = 'public'
4 ORDER BY table_name;
```

Data Output Messages Notifications

	table_name	column_name	is_nullable
	name	name	character varying (3)
47	flights	sch_departure_time	NO
48	flights	sch_arrival_time	NO
49	flights	departing_airport_id	NO
50	flights	arriving_airport_id	NO
51	flights	airline_id	NO
52	flights	act_departure_time	NO
53	flights	act_arrival_time	NO
54	flights	created_at	NO
55	flights	updated_at	NO
56	flights	departing_gate	NO
57	passengers	passenger_id	NO
58	passengers	gender	NO
59	passengers	country_of_citizens...	NO
60	passengers	country_of_residence	NO
61	passengers	passport_number	NO
62	passengers	updated_at	NO
63	passengers	created_at	NO
64	passengers	date_of_birth	NO
65	passengers	first_name	NO
66	passengers	last_name	NO
67	security_che...	created_at	NO
68	security_che...	passenger_id	NO
69	security_che...	updated_at	NO
70	security_che...	check_result	NO
71	security_che...	security_check_id	NO

Total rows: 71 Query complete 00:00:00.278

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Dashboard Properties SQL Statistics Dependencies Dependents Processes airport/postgres@PostgreSQL 18*

Query Query History

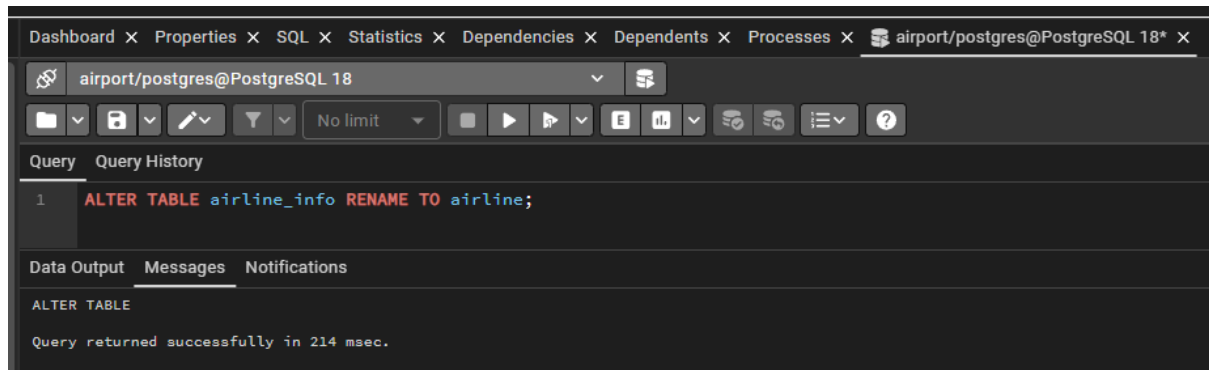
```
1 SELECT table_name, constraint_name FROM information_schema.table_constraints
2 WHERE constraint_type = 'PRIMARY KEY' AND table_schema = 'public';
```

Data Output Messages Notifications

	table_name	constraint_name
	name	name
1	airline_info	airline_info_pkey
2	airport	airport_pkey
3	passengers	passengers_pkey
4	flights	flights_pkey
5	booking	booking_pkey
6	booking_flight	booking_flight_pkey
7	baggage	baggage_pkey
8	baggage_ch...	baggage_check_p...
9	boarding_pa...	boarding_pass_pk...
10	security_che...	security_check_pk...

Total rows: 10 Query complete 00:00:00.224

Task 4



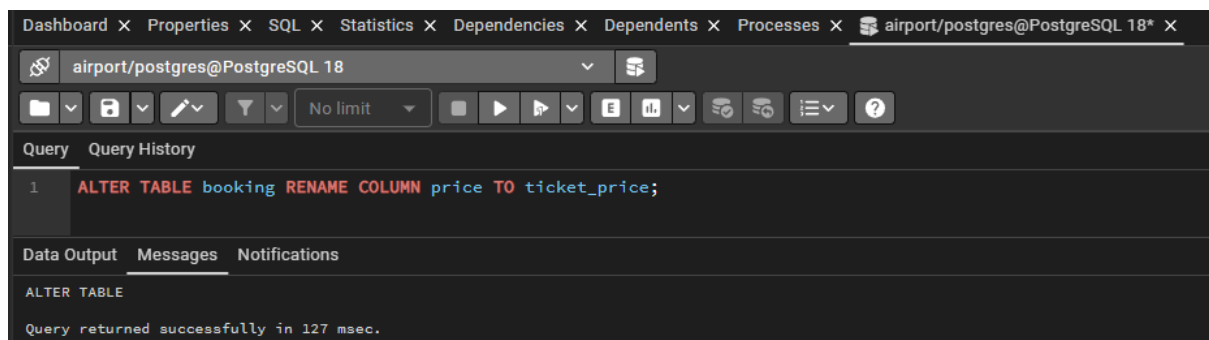
The screenshot shows the DBeaver SQL editor interface. The top toolbar includes icons for file operations, query execution, and settings. The query editor contains the following SQL statement:

```
1 ALTER TABLE airline_info RENAME TO airline;
```

The 'Messages' tab is active, displaying the output:

```
ALTER TABLE  
Query returned successfully in 214 msec.
```

Task 5



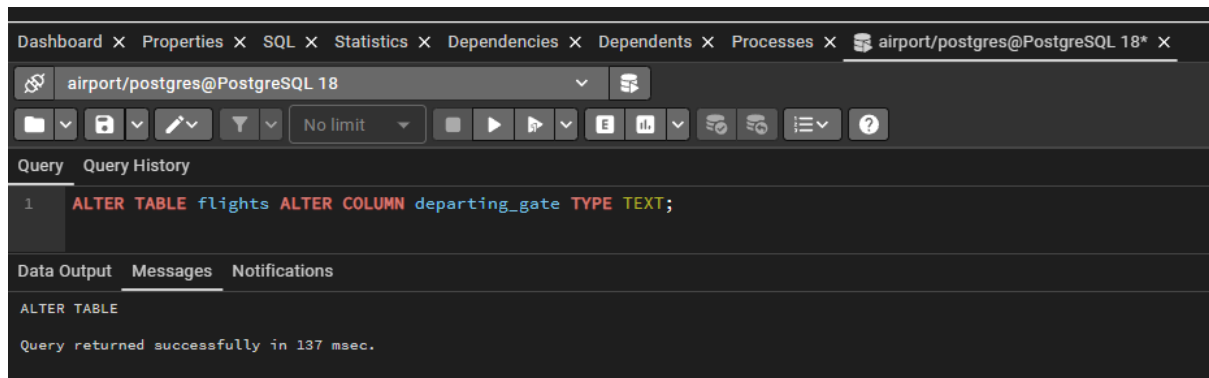
The screenshot shows the DBeaver SQL editor interface. The query editor contains the following SQL statement:

```
1 ALTER TABLE booking RENAME COLUMN price TO ticket_price;
```

The 'Messages' tab is active, displaying the output:

```
ALTER TABLE  
Query returned successfully in 127 msec.
```

Task 6



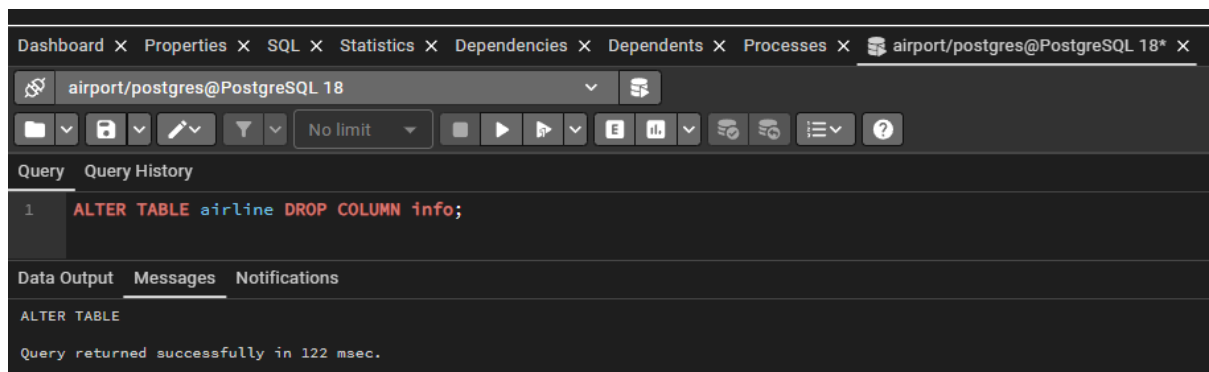
The screenshot shows the DBeaver SQL editor interface. The query editor contains the following SQL statement:

```
1 ALTER TABLE flights ALTER COLUMN departing_gate TYPE TEXT;
```

The 'Messages' tab is active, displaying the output:

```
ALTER TABLE  
Query returned successfully in 137 msec.
```

Task 7



The screenshot shows the DBeaver SQL editor interface. The query editor contains the following SQL statement:

```
1 ALTER TABLE airline DROP COLUMN info;
```

The 'Messages' tab is active, displaying the output:

```
ALTER TABLE  
Query returned successfully in 122 msec.
```

Task 8

Passengers with Secutiry_check, Booking, Baggage_check by passenger_id;

Dashboard X Properties X SQL X Statistics X Dependencies X Dependents X Processes X airport/postgres@PostgreSQL 18* X

airport/postgres@PostgreSQL 18

No limit

Query Query History

```
1 ALTER TABLE security_check ADD CONSTRAINT fk_security_passenger
2   FOREIGN KEY (passenger_id) REFERENCES passengers(passenger_id);
3 ALTER TABLE booking ADD CONSTRAINT fk_booking_passenger
4   FOREIGN KEY (passenger_id) REFERENCES passengers(passenger_id);
5 ALTER TABLE baggage_check ADD CONSTRAINT fk_bagcheck_passenger
6   FOREIGN KEY (passenger_id) REFERENCES passengers(passenger_id);
```

Data Output Messages Notifications

ALTER TABLE

Query returned successfully in 132 msec.

Booking with Baggage_check, Baggage, Boarding_pass, Booking_flight by booking_id;

Dashboard X Properties X SQL X Statistics X Dependencies X Dependents X Processes X airport/postgres@PostgreSQL 18* X

airport/postgres@PostgreSQL 18

No limit

Query Query History

```
1 ALTER TABLE baggage_check ADD CONSTRAINT fk_bagcheck_booking
2   FOREIGN KEY (booking_id) REFERENCES booking(booking_id);
3 ALTER TABLE baggage ADD CONSTRAINT fk_baggage_booking
4   FOREIGN KEY (booking_id) REFERENCES booking(booking_id);
5 ALTER TABLE boarding_pass ADD CONSTRAINT fk_boarding_booking
6   FOREIGN KEY (booking_id) REFERENCES booking(booking_id);
7 ALTER TABLE booking_flight ADD CONSTRAINT fk_bf_booking
8   FOREIGN KEY (booking_id) REFERENCES booking(booking_id);
```

Data Output Messages Notifications

ALTER TABLE

Query returned successfully in 228 msec.

Flights with Booking_flight by flight_id;

Dashboard X Properties X SQL X Statistics X Dependencies X Dependents X Processes X airport/postgres@PostgreSQL 18* X

airport/postgres@PostgreSQL 18

No limit

Query Query History

```
1 ALTER TABLE booking_flight ADD CONSTRAINT fk_bf_flight
2   FOREIGN KEY (flight_id) REFERENCES flights(flight_id);
```

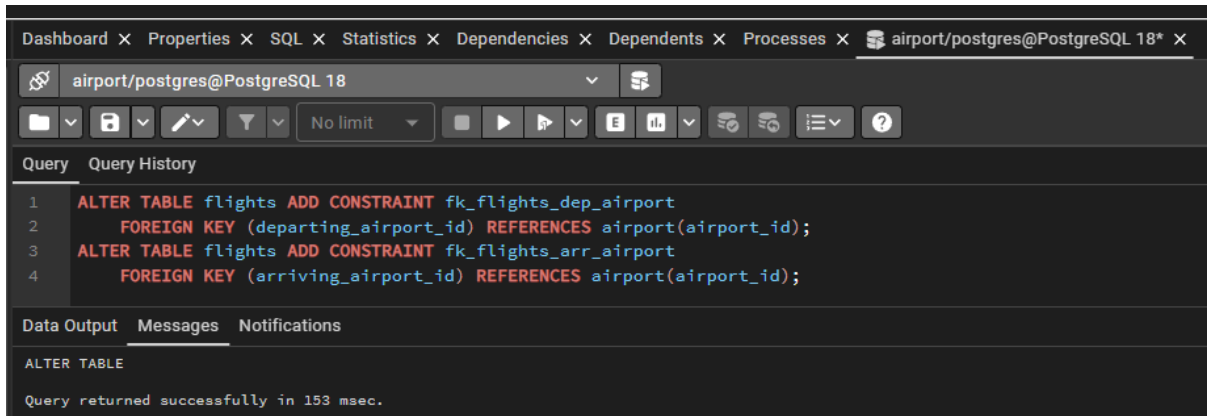
Data Output Messages Notifications

ALTER TABLE

Query returned successfully in 132 msec.

Airport with Flights by departing_airport_id;

Airport with Flights by arriving_airport_id;



The screenshot shows a PostgreSQL client interface with a dark theme. The top navigation bar includes tabs for Dashboard, Properties, SQL, Statistics, Dependencies, Dependents, and Processes. The active tab is 'airport/postgres@PostgreSQL 18'. Below the navigation bar is a toolbar with icons for file operations, query execution, and settings. The main area is divided into two sections: 'Query' and 'Query History'. The 'Query' section contains four lines of SQL code:
1 ALTER TABLE flights ADD CONSTRAINT fk_flights_dep_airport
2 FOREIGN KEY (departing_airport_id) REFERENCES airport(airport_id);
3 ALTER TABLE flights ADD CONSTRAINT fk_flights_arr_airport
4 FOREIGN KEY (arriving_airport_id) REFERENCES airport(airport_id);
The 'Query History' section shows the execution of the first line of the query, with the message 'Query returned successfully in 153 msec.'

```
Dashboard X Properties X SQL X Statistics X Dependencies X Dependents X Processes X airport/postgres@PostgreSQL 18 X
```

airport/postgres@PostgreSQL 18

No limit

Query Query History

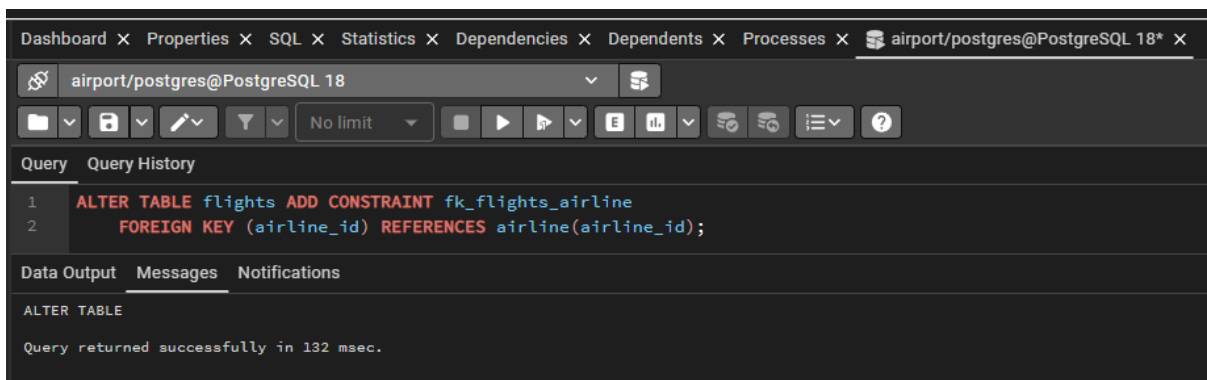
```
1 ALTER TABLE flights ADD CONSTRAINT fk_flights_dep_airport
2 FOREIGN KEY (departing_airport_id) REFERENCES airport(airport_id);
3 ALTER TABLE flights ADD CONSTRAINT fk_flights_arr_airport
4 FOREIGN KEY (arriving_airport_id) REFERENCES airport(airport_id);
```

Data Output Messages Notifications

ALTER TABLE

Query returned successfully in 153 msec.

Airline with Flights by airline_id;



The screenshot shows a PostgreSQL client interface with a dark theme. The top navigation bar includes tabs for Dashboard, Properties, SQL, Statistics, Dependencies, Dependents, and Processes. The active tab is 'airport/postgres@PostgreSQL 18'. Below the navigation bar is a toolbar with icons for file operations, query execution, and settings. The main area is divided into two sections: 'Query' and 'Query History'. The 'Query' section contains two lines of SQL code:
1 ALTER TABLE flights ADD CONSTRAINT fk_flights_airline
2 FOREIGN KEY (airline_id) REFERENCES airline(airline_id);
The 'Query History' section shows the execution of the first line of the query, with the message 'Query returned successfully in 132 msec.'

```
Dashboard X Properties X SQL X Statistics X Dependencies X Dependents X Processes X airport/postgres@PostgreSQL 18 X
```

airport/postgres@PostgreSQL 18

No limit

Query Query History

```
1 ALTER TABLE flights ADD CONSTRAINT fk_flights_airline
2 FOREIGN KEY (airline_id) REFERENCES airline(airline_id);
```

Data Output Messages Notifications

ALTER TABLE

Query returned successfully in 132 msec.

Dashboard X Properties X SQL X Statistics X Dependencies X Dependents X Processes X airport/postgres@PostgreSQL 18* X

airport/postgres@PostgreSQL 18

Query Query History

```

1 SELECT tc.table_name, tc.constraint_name, kcu.column_name,
2       ccu.table_name AS ref_table, ccu.column_name AS ref_column
3 FROM information_schema.table_constraints tc
4 JOIN information_schema.key_column_usage kcu
5   ON tc.constraint_name = kcu.constraint_name
6 JOIN information_schema.constraint_column_usage ccu
7   ON tc.constraint_name = ccu.constraint_name
8 WHERE tc.constraint_type = 'FOREIGN KEY' AND tc.table_schema = 'public'
9 ORDER BY tc.table_name;

```

Data Output Messages Notifications

	table_name name	constraint_name name	column_name name	ref_table name	ref_column name
1	baggage	fk_baggage_booki...	booking_id	booking	booking_id
2	baggage_ch...	fk_bagcheck_pass...	passenger_id	passenge...	passenger_id
3	baggage_ch...	fk_bagcheck_book...	booking_id	booking	booking_id
4	boarding_pa...	fk_boarding_booki...	booking_id	booking	booking_id
5	booking	fk_booking_passe...	passenger_id	passenge...	passenger_id
6	booking_flight	fk_bf_flight	flight_id	flights	flight_id
7	booking_flight	fk_bf_booking	booking_id	booking	booking_id
8	flights	fk_flights_airline	airline_id	airline	airline_id
9	flights	fk_flights_dep_air...	departing_airport...	airport	airport_id
10	flights	fk_flights_arr_airp...	arriving_airport_id	airport	airport_id
11	security_che...	fk_security_passe...	passenger_id	passenge...	passenger_id

NORMALIZATION SECTION

Task 1

The table violates 1NF, and therefore 2NF and 3NF as well. The column `seat_numbers` stores a comma-separated list ("12A, 12B"), so its values are not atomic. `passenger_full_name` is also a composite value (first + last name). Even after fixing atomicity, the table contains partial and transitive dependencies: `booking_id` → `flight_number` → `airline_name`, `departure_airport_name` → `departure_city`, `passenger_passport_number` → `passenger_full_name`. This causes redundancy and update, insert and delete anomalies

Task 2

1NF — one row per seat:

Booking_receipt(**booking_id**, **seat_number**, passenger_full_name, passenger_passport_number, flight_number, departure_airport_name, departure_city, arrival_airport_name, arrival_city, airline_name, ticket_price)

2NF — all attributes except the seat depend only on `booking_id` (part of the key), so they are moved out:

Booking(**booking_id**, passenger_full_name, passenger_passport_number,

flight_number, departure_airport_name, departure_city, arrival_airport_name, arrival_city, airline_name, ticket_price)

Booking_seat(**booking_id** (FK), **seat_number**)

3NF — transitive dependencies are removed:

Passenger(**passenger_passport_number**, passenger_full_name)

Airport(**airport_name**, city)

Airline(**airline_name**)

Flight(**flight_number**, departure_airport_name (FK), arrival_airport_name (FK), airline_name (FK))

Booking(**booking_id**, passenger_passport_number (FK), flight_number (FK), ticket_price)

Booking_seat(**booking_id** (FK), **seat_number**)

Task 3

1NF: seat_numbers is multi-valued, so it is split into one row per seat.

2NF (partial dependency on the composite key): booking_id →

passenger_passport_number, flight_number, ticket_price

3NF (transitive dependencies):

passenger_passport_number → passenger_full_name

flight_number → departure_airport_name, arrival_airport_name, airline_name

departure_airport_name → departure_city

arrival_airport_name → arrival_city

Task 4

The screenshot shows a PostgreSQL database interface with a query window and a data output window. The query window contains the following SQL query:

```
1 SELECT column_name, data_type FROM information_schema.columns
2 WHERE table_name = 'flights';
```

The data output window displays the results of the query, showing the column names and data types for the 'flights' table. The results are as follows:

	column_name	data_type
	name	character varying
1	flight_id	integer
2	sch_departure_ti...	timestamp without time zo...
3	sch_arrival_time	timestamp without time zo...
4	departing_airport...	integer
5	arriving_airport_id	integer
6	created_at	timestamp without time zo...
7	updated_at	timestamp without time zo...
8	airline_id	integer
9	act_departure_time	timestamp without time zo...
10	act_arrival_time	timestamp without time zo...
11	departing_gate	text
12	arriving_gate	character varying

Primary key: flight_id

Functional dependencies: $\text{flight_id} \rightarrow \text{sch_departure_time}, \text{sch_arrival_time}, \text{departing_airport_id}, \text{arriving_airport_id}, \text{departing_gate}, \text{arriving_gate}, \text{airline_id}, \text{act_departure_time}, \text{act_arrival_time}, \text{created_at}, \text{updated_at}$.

There are no transitive dependencies. $\text{departing_airport_id}$, $\text{arriving_airport_id}$ and airline_id are foreign keys, and no other non-key attribute of Flights depends on them: airport name, city and country are stored in Airport, and airline data is stored in Airline. The gate is not determined by the airport alone, because an airport has many gates and the gate depends on the specific flight. Therefore, Flights is in 3NF. It would violate 3NF if, for example, departing_city were added to Flights ($\text{flight_id} \rightarrow \text{departing_airport_id} \rightarrow \text{departing_city}$)

Task 5

pgAdmin 4

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airtpostgres@PostgreSQL 18*

Query

```

1 CREATE SCHEMA norm;
2 CREATE TABLE norm.airline (
3     airline_name VARCHAR(50) NOT NULL PRIMARY KEY
4 );
5 CREATE TABLE norm.airport (
6     airport_name VARCHAR(50) NOT NULL PRIMARY KEY,
7     city VARCHAR(50) NOT NULL
8 );
9 CREATE TABLE norm.flight (
10    flight_number VARCHAR(10) NOT NULL PRIMARY KEY,
11    departure_airport_name VARCHAR(50) NOT NULL,
12    arrival_airport_name VARCHAR(50) NOT NULL,
13    airline_name VARCHAR(50) NOT NULL,
14    CONSTRAINT fk_n_flight_dep FOREIGN KEY (departure_airport_name) REFERENCES norm.airport(airport_name),
15    CONSTRAINT fk_n_flight_arr FOREIGN KEY (arrival_airport_name) REFERENCES norm.airport(airport_name),
16    CONSTRAINT fk_n_flight_airline FOREIGN KEY (airline_name) REFERENCES norm.airline(airline_name)
17 );
18 CREATE TABLE norm.passenger (
19    passenger_passport_number VARCHAR(20) NOT NULL PRIMARY KEY,
20    passenger_full_name VARCHAR(100) NOT NULL
21 );
22 CREATE TABLE norm.booking (
23    booking_id INT NOT NULL PRIMARY KEY,
24    passenger_passport_number VARCHAR(20) NOT NULL,
25    flight_number VARCHAR(10) NOT NULL,
26    ticket_price DECIMAL(7,2) NOT NULL,
27    CONSTRAINT fk_n_booking_pass FOREIGN KEY (passenger_passport_number) REFERENCES norm.passenger(passenger_passport_number),
28    CONSTRAINT fk_n_booking_flight FOREIGN KEY (flight_number) REFERENCES norm.flight(flight_number)
29 );
30 CREATE TABLE norm.booking_seat (
31    booking_id INT NOT NULL,
32    seat_number VARCHAR(5) NOT NULL,
33    PRIMARY KEY (booking_id, seat_number),
34    CONSTRAINT fk_n_seat_booking FOREIGN KEY (booking_id) REFERENCES norm.booking(booking_id)
35 );
36 CREATE TABLE norm.booking_receipt_raw (
37    booking_id INT, passenger_full_name VARCHAR(100), passenger_passport_number VARCHAR(20),
38    flight_number VARCHAR(10), departure_airport_name VARCHAR(50), departure_city VARCHAR(50),
39    arrival_airport_name VARCHAR(50), arrival_city VARCHAR(50), airline_name VARCHAR(50),
40    seat_numbers VARCHAR(50), ticket_price DECIMAL(7,2)
41 );

```

Data Output Messages Notifications

CREATE TABLE

Query returned successfully in 194 msec.

Total rows: Query complete 00:00:00.194

pgAdmin 4

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Query Editor

airport/postgres@PostgreSQL 18

Query History

```
1 INSERT INTO norm.booking_receipt_raw VALUES
2 (1, 'Aidar Nurlanov', 'N1234567', 'KC981', 'Almaty International', 'Almaty', 'Astana International', 'Astana', 'Air Astana', '12A, 12B', 85000.00),
3 (2, 'Dena Seitova', 'N2345678', 'KC981', 'Almaty International', 'Almaty', 'Astana International', 'Astana', 'Air Astana', '14C', 42500.00),
4 (3, 'Ivan Petrov', 'N3456789', 'DV715', 'Pavlodar Airport', 'Pavlodar', 'Almaty International', 'Almaty', 'SCAT', '5D', 38000.00),
5 (4, 'Aidar Nurlanov', 'N1234567', 'KC837', 'Astana International', 'Astana', 'Istanbul Airport', 'Istanbul', 'Air Astana', '20A, 20B, 20C', 390000.00),
6 (5, 'Madina Zhaksy', 'N4567890', 'FS7231', 'Astana International', 'Astana', 'Pavlodar Airport', 'Pavlodar', 'FlyArystan', '3F', 19900.00),
7 (6, 'Ivan Petrov', 'N3456789', 'KC981', 'Almaty International', 'Almaty', 'Astana International', 'Astana', 'Air Astana', '7A, 7B', 84000.00);
8
9 SELECT * FROM norm.booking_receipt_raw;
```

Data Output

	booking_id integer	passenger_full_name character varying (100)	passenger_passport_number character varying (20)	flight_number character varying (10)	departure_airport_name character varying (50)	departure_city character varying (50)	arrival_airport_name character varying (50)	arrival_city character varying (50)	airline_name character varying (50)	seat_numbers character varying (50)	ticket_price numeric (10,2)
1	1	Aidar Nurlanov	N1234567	KC901	Almaty International	Almaty	Astana International	Astana	Air Astana	12A, 12B	85000.00
2	2	Dena Seitova	N2345678	KC901	Almaty International	Almaty	Astana International	Astana	Air Astana	14C	42500.00
3	3	Ivan Petrov	N3456789	DV715	Pavlodar Airport	Pavlodar	Almaty International	Almaty	SCAT	5D	38000.00
4	4	Aidar Nurlanov	N1234567	KC037	Astana International	Astana	Istanbul Airport	Istanbul	Air Astana	20A, 20B, 20C	390000.00
5	5	Madina Zhaksy	N4567890	FS7231	Astana International	Astana	Pavlodar Airport	Pavlodar	FlyArystan	3F	19900.00
6	6	Ivan Petrov	N3456789	KC901	Almaty International	Almaty	Astana International	Astana	Air Astana	7A, 7B	84000.00

Total rows: 6 Query complete 00:00:00.165

Query Editor

airport/postgres@PostgreSQL 18

Query

```
1 -- INSERT INTO norm.airline (airline_name)
2 SELECT DISTINCT airline_name FROM norm.booking_receipt_raw;
3
4 -- -- INSERT INTO norm.airport (airport_name, city)
5 -- SELECT departure_airport_name, departure_city FROM norm.booking_receipt_raw UNION
6 -- SELECT arrival_airport_name, arrival_city FROM norm.booking_receipt_raw;
7
8 -- -- INSERT INTO norm.flight
9 -- SELECT DISTINCT flight_number, departure_airport_name, arrival_airport_name, airline_name FROM norm.booking_receipt_raw;
10
11 -- -- INSERT INTO norm.passenger
12 -- SELECT DISTINCT passenger_passport_number, passenger_full_name FROM norm.booking_receipt_raw;
13
14 -- -- INSERT INTO norm.booking
15 -- SELECT booking_id, passenger_passport_number, flight_number, ticket_price FROM norm.booking_receipt_raw;
16
17 -- -- INSERT INTO norm.booking_seat (booking_id, seat_number)
18 -- SELECT booking_id, TRIM(s) FROM norm.booking_receipt_raw, unnest(string_to_array(seat_numbers, ',')) AS s;
```

Data Output

	airline_name character varying (50)
1	SCAT
2	FlyArystan

Query Editor

airport/postgres@PostgreSQL 18

Query

```
1 -- INSERT INTO norm.airline (airline_name)
2 -- SELECT DISTINCT airline_name FROM norm.booking_receipt_raw;
3
4 -- -- INSERT INTO norm.airport (airport_name, city)
5 SELECT departure_airport_name, departure_city FROM norm.booking_receipt_raw UNION SELECT arrival_airport_name, arrival_city FROM norm.booking_receipt_raw;
6
7 -- -- INSERT INTO norm.flight
8 -- SELECT DISTINCT flight_number, departure_airport_name, arrival_airport_name, airline_name FROM norm.booking_receipt_raw;
9
10 -- -- INSERT INTO norm.passenger
11 -- SELECT DISTINCT passenger_passport_number, passenger_full_name FROM norm.booking_receipt_raw;
12
13 -- -- INSERT INTO norm.booking
14 -- SELECT booking_id, passenger_passport_number, flight_number, ticket_price FROM norm.booking_receipt_raw;
15
16 -- -- INSERT INTO norm.booking_seat (booking_id, seat_number)
17 -- SELECT booking_id, TRIM(s) FROM norm.booking_receipt_raw, unnest(string_to_array(seat_numbers, ',')) AS s;
```

Data Output

	departure_airport_name character varying (50)	departure_city character varying (50)
1	Pavlodar Airport	Pavlodar
2	Astana International	Astana
3	Almaty International	Almaty
4	Istanbul Airport	Istanbul

Dashboard × Properties × SQL × Statistics × Dependencies × Dependents × Processes × airport/postgres@PostgreSQL 18* ×

airport/postgres@PostgreSQL 18

No limit

Query Query History

```
1 -- INSERT INTO norm.airline (airline_name)
2 -- SELECT DISTINCT airline_name FROM norm.booking_receipt_raw;
3
4 -- -- INSERT INTO norm.airport (airport_name, city)
5 -- SELECT departure_airport_name, departure_city FROM norm.booking_receipt_raw UNION SELECT arrival_airport_name, arrival_city FROM norm.booking_receipt_raw;
6
7 -- -- INSERT INTO norm.flight
8 SELECT DISTINCT flight_number, departure_airport_name, arrival_airport_name, airline_name FROM norm.booking_receipt_raw;
9
10 -- -- INSERT INTO norm.passenger
11 -- SELECT DISTINCT passenger_passport_number, passenger_full_name FROM norm.booking_receipt_raw;
12
13 -- -- INSERT INTO norm.booking
14 -- SELECT booking_id, passenger_passport_number, flight_number, ticket_price FROM norm.booking_receipt_raw;
15
16 -- -- INSERT INTO norm.booking_seat (booking_id, seat_number)
17 -- SELECT booking_id, TRIM(s) FROM norm.booking_receipt_raw, unnest(string_to_array(seat_numbers, ',')) AS s;
```

Data Output Messages Notifications

	flight_number character varying (10)	departure_airport_name character varying (50)	arrivalAirport_name character varying (50)	airline_name character varying (50)
1	FS7231	Astana International	Pavlodar Airport	FlyArystan
2	KC037	Astana International	Istanbul Airport	Air Astana
3	KC901	Almaty International	Astana International	Air Astana
4	DV715	Pavlodar Airport	Almaty International	SCAT

Dashboard × Properties × SQL × Statistics × Dependencies × Dependents × Processes × airport/postgres@PostgreSQL 18* ×

airport/postgres@PostgreSQL 18

No limit

Query Query History

```
1 -- INSERT INTO norm.airline (airline_name)
2 -- SELECT DISTINCT airline_name FROM norm.booking_receipt_raw;
3
4 -- -- INSERT INTO norm.airport (airport_name, city)
5 -- SELECT departure_airport_name, departure_city FROM norm.booking_receipt_raw UNION SELECT arrival_airport_name, arrival_city FROM norm.booking_receipt_raw;
6
7 -- -- INSERT INTO norm.flight
8 -- SELECT DISTINCT flight_number, departure_airport_name, arrival_airport_name, airline_name FROM norm.booking_receipt_raw;
9
10 -- -- INSERT INTO norm.passenger
11 SELECT DISTINCT passenger_passport_number, passenger_full_name FROM norm.booking_receipt_raw;
12
13 -- -- INSERT INTO norm.booking
14 -- SELECT booking_id, passenger_passport_number, flight_number, ticket_price FROM norm.booking_receipt_raw;
15
16 -- -- INSERT INTO norm.booking_seat (booking_id, seat_number)
17 -- SELECT booking_id, TRIM(s) FROM norm.booking_receipt_raw, unnest(string_to_array(seat_numbers, ',')) AS s;
```

Data Output Messages Notifications

	passenger_passport_number character varying (20)	passenger_full_name character varying (100)
1	N1234567	Aidar Nurlanov
2	N3456789	Ivan Petrov
3	N4567890	Madina Zhaksy
4	N2345678	Dana Seitova

Dashboard X Properties X SQL X Statistics X Dependencies X Dependents X Processes X airport/postgres@PostgreSQL 18* X

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Query Query History

```

1  -- INSERT INTO norm.airline (airline_name)
2  -- SELECT DISTINCT airline_name FROM norm.booking_receipt_raw;
3
4  -- -- INSERT INTO norm.airport (airport_name, city)
5  -- SELECT departure_airport_name, departure_city FROM norm.booking_receipt_raw UNION SELECT arrival_airport_name,
6
7  -- -- INSERT INTO norm.flight
8  -- SELECT DISTINCT flight_number, departure_airport_name, arrival_airport_name, airline_name FROM norm.booking_re
9
10 -- -- INSERT INTO norm.passenger
11 -- SELECT DISTINCT passenger_passport_number, passenger_full_name FROM norm.booking_receipt_raw;
12
13 -- -- INSERT INTO norm.booking
14 -- SELECT booking_id, passenger_passport_number, flight_number, ticket_price FROM norm.booking_receipt_raw;
15
16 -- -- INSERT INTO norm.booking_seat (booking_id, seat_number)
17 SELECT booking_id, TRIM(s) FROM norm.booking_receipt_raw, unnest(string_to_array(seat_numbers, ',')) AS s;

```

Data Output Messages Notifications

	booking_id integer	btrim text
1	1	12A
2	1	12B
3	2	14C
4	3	5D
5	4	20A
6	4	20B
7	4	20C
8	5	3F
9	6	7A
10	6	7B

Dashboard X Properties X SQL X Statistics X Dependencies X Dependents X Processes X airport/postgres@PostgreSQL 18* X

airport/postgres@PostgreSQL 18

Query Query History

```

1  SELECT b.booking_id, p.passenger_full_name, f.flight_number,
2         da.city AS departure_city, aa.city AS arrival_city, f.airline_name,
3         string_agg(s.seat_number, ',' ORDER BY s.seat_number) AS seat_numbers,
4         b.ticket_price
5  FROM norm.booking b
6       JOIN norm.passenger p ON p.passenger_passport_number = b.passenger_passport_number
7       JOIN norm.flight f ON f.flight_number = b.flight_number
8       JOIN norm.airport da ON da.airport_name = f.departure_airport_name
9       JOIN norm.airport aa ON aa.airport_name = f.arrival_airport_name
10      JOIN norm.booking_seat s ON s.booking_id = b.booking_id
11  GROUP BY b.booking_id, p.passenger_full_name, f.flight_number, da.city, aa.city,
12         f.airline_name, b.ticket_price
13  ORDER BY b.booking_id;

```

Data Output Messages Notifications

	booking_id integer	passenger_full_name character varying (100)	flight_number character varying (10)	departure_city character varying (50)	arrival_city character varying (50)	airline_name character varying (50)	seat_numbers text	ticket_price numeric (10,2)
1	1	Aidar Nurlanov	KC901	Almaty	Astana	Air Astana	12A, 12B	85000.00
2	2	Dana Seitova	KC901	Almaty	Astana	Air Astana	14C	42500.00
3	3	Ivan Petrov	DV715	Pavlodar	Almaty	SCAT	5D	38000.00
4	4	Aidar Nurlanov	KC037	Astana	Istanbul	Air Astana	20A, 20B, 20C	390000.00
5	5	Madina Zhaksy	FS7231	Astana	Pavlodar	FlyArystan	3F	19900.00
6	6	Ivan Petrov	KC901	Almaty	Astana	Air Astana	7A, 7B	84000.00

The JOIN query reconstructs the original receipt, which shows that no information was lost during normalization